Christopher Smeriglio

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Software Engineer with 7 years of experience in development and test engineering and a strong passion for embedded and hardware-centric systems. I bring hands-on experience with embedded bring-up and complex test environments, and I specialize in designing and implementing robust, scalable automated test frameworks. Known as a flexible, creative problem-solver, I consistently step outside conventional boundaries to deliver effective solutions. I pair strong technical leadership with exceptional interpersonal skills, frequently breaking down silos and driving cross-functional alignment. I have extensive experience presenting technical strategies, risks, and outcomes to executive leadership and securing buy-in to move initiatives forward. A lifelong learner and mentor, I actively support junior engineers through onboarding and early-career development.

Technical Skills

- **Top Skills: Python, C# (.NET), Test Automation, Gherkin, Behavior Driven Development, GIT, Sprint Paradigm**
- Programming Languages: Python, C#, C++, C, SQL, Java, Multisim, ARM, SCPI, MATLAB
- Hardware: Optical Tracking, Depth Imaging, Embedded SoM, Robotics, Oscilloscope, Waveform Generator
- Software Tools: Linux (Yocto), Jenkins, BitBucket, Docker, SSH, GNU Bash, NI Max, Ensenso, URCap, LabVIEW
- Software Methodologies: Object-Oriented Programming, Dependency Injection, Agile, MVVM, SDLC, SOLID
- Quality Skills: CAPA, FMEA, Requirement Definition, Regulative Standards, Root Cause Analysis, Data Analytics

Education

August 2015 – August 2019

Wentworth Institute of Technology - Boston, MA — *Bachelor of Science in Computer Engineering*

Professional Experience

Proprio – Medtech, Med Device — Seattle, WA

BE Lead Software Development Engineer in Test — August 2023 – February 2026

- Operated in a regulated Medical Device release cycle including working with the FDA, MDSAP, IEC & ISO
- Created automated tool to update requirements in Word document-based protocols utilizing JAMA's API
- Oversaw test protocols, work instructions, and procedures to ensure consistent quality for a medical device
- Managed in-house systems configuration and provided support when critical systems went down
- Collaborated with hardware and clinical teams to ensure optimal user experience and improve patient outcome
- Defined processes for software deployment at critical hospital sites
- Functioned as sole test team member and scaled test team to better meet the company needs

Select Achievements:

- Integrated automation test suite based on Gherkin in Python to enhance Behavior Driven Development
- Overhauled Embedded test infrastructure to significantly increase code coverage to over 90%
- Pitched projects to executives and obtained critical buy-in
- Developed monitoring tools to detect memory leaks, identify bottlenecks and optimize system utilization
- Lead weekly testing meetings to determine priorities, discuss new issues and assist team members

Kymeta Corp – SatCom, TeleCom — Redmond, WA

Test Engineering Development Team; *Software Engineer* — January 2021 – August 2023

- Utilized SCPI commands to automate testing procedures and optimized for maximum efficiency
- Reduced manufacturing costs by reducing test times through optimization and removal of external software
- Integrated scrum and sprint paradigm into workflow to optimize productivity in a collaborative work environment
- Maintained and supported infrastructure to manage and regularly test code set to ensure a stable product
- Applied knowledge of signal processing and S parameters to utilize network analyzers for complex testing

Select Achievements:

- Collaborated to design and implement new test software for an external production environment
- Applied C# and Python to develop large tool sets for testing complex antenna system
- Developed algorithms to parse large data sets and extract relevant data to verify hardware and support R&D efforts
- Integrated dependency injection, MVVM and git submodules to simplify complex code base

Microsoft Cold Logic - Quantum Computing — Redmond, WA

Test Lab Engineer — February 2020 – January 2021

- Applied superconducting concepts to aid in development of digital circuits using quantum mechanics
- Programmed Python scripts to process and interpret RF characterization data using knowledge of waveforms
- Utilized VNA and TDR to troubleshoot and verify chip designs met specification
- Designed and implemented automation of data entry of chip metadata using Python and SQL databases
- Oversaw development of Dynamics365 inventory management system and designed implementation for lab
- Collaborated with design engineers to assess/correct design flaws related to RF characterization, BERT, SERDES

Cytonome - Biotech — Bedford, MA

Systems Software Engineer — August 2018 – July 2019

- Programmed automation scripts to automate software testing procedures to be utilized by quality assurance team
- Worked with precise lab equipment in a regulated environment to test and develop accurate medical cell-sorter
- Identified and repaired failing components in fluidics and electrical systems in research lab cell-sorters
- Built file parsers in Python to extract essential information from log files for use by fellow systems team members

Harvard University Information Technology — Cambridge, MA

Technical Support Engineer — May 2017 – September 2017

- Provided hardware and software technical support to Harvard University deans, faculty, and fellows
- Deployed hardware implementation plans for overhauled training suites for Harvard Library Technology Services

Additional Projects

Artemis Server

- Implemented Linux-based server for media streaming, AI Upscaling, internal web hosting, and network tooling
- Applied Knowledge of Docker containers, GNU Bash scripts, and Python scripting to develop automation
- Permission management, power plans, and automated startup tasks

Computer Building

- Built and maintain a network of high-performance desktops across operating systems with varying hardware
- Familiar with optimizing components and working with sensitive, expensive electronics
- Extensive knowledge of Linux, Proton, Integrating Compatibility layers, Boot Loaders (GRUB)
- Modifying and Flashing BIOS images to personalize my PC

Automated Highway Lane Allocator

- Simulated professional product development process: concept design, business proposal, consumer feedback, large-scale implementation feasibility, safety and ethics evaluations, modelling and presentation of final product with a development team of students and a faculty member serving as a consultant
- Developed simulation of two-way traffic flow and automated lane allocation algorithm to shift lane directionality to improve throughput in response to traffic
- Designed and manufactured prototype traffic detection system using Arduino, pressure sensors, and basic tubing
- Conducted regular professional status update meetings and logged project progress in an engineering notebook

Verilog Pipeline

- Implemented Verilog code to recreate the components of a fully functional single-core processor to run myriad instructions in ARM machine code

Analog Audio Equalizer Circuit

- Designed a complex circuit using 6 op-amps: a buffer, low-pass, high-pass, and band-pass filters, a voltage adder to combine the filtered signals, and a current amplifier to improve output signal
- Employed circuit as an analog equalizer, filtering high and low frequencies from input audio signals